Nomenclature

State and control. $x \in \mathbb{R}^d$ is the state (inventory, and the signal when one is modelled); $u \in \mathbb{R}^m$ is the control (trading rate); d is the state dimension and m the control dimension, with $m = p$ and $d = p$ in the inventory-only benchmark; p is the number of assets.

Problem data. T is the horizon; $Q = \lambda\sigma^2 I$ the inventory-risk penalty; $R = \eta I$ the temporary market impact; Q_T the terminal inventory penalty; $\Sigma = \sigma I$ the diffusion; A, B the dynamics; λ risk aversion (a preference, not measurable); σ relative volatility; η the impact coefficient, identified with half the Roll effective spread; $k = \sqrt{Q/\eta}$ the liquidation rate.

Solution. $V(x,t) = -\frac{1}{2}x^\top P(t)x + c(t)$ is the value function; $P(t)$ solves the matrix Riccati equation with $P(T) = Q_T$; $u^\star = -R^{-1}B^\top P(t)x$ is the optimal feedback; $J(u)$ is the objective under control u.

Discretisation. h is the mesh spacing, N the number of grid points per axis, $N^\star$ the smallest N reaching the stated accuracy target; R_h is the reconstruction that turns nodal values into a continuous function, of polynomial degree k; $\nabla_h V$ is the gradient recovered from it; ε is the finite-difference probe step, taken as $\varepsilon = c\,h^\alpha$ when the scaling is being tested.

Errors. All errors are relative L^2 norms, $\|a - a^\star\|_2/\|a^\star\|_2$; every convergence order is the least-squares slope of log(error) against $\log h$; R^2 is reported as a secondary, dimensionless accuracy number and is never used to fit an order.

Measured configurations. Default coefficients $\eta = \lambda\sigma^2 = 0.01$, $Q_T = 0.1$, $\sigma = 0.01$, $T = 1$ (giving $k = 1$); accuracy-matched grid resolution $N^\star = 21$ at every dimension for the quadratic benchmark; CPU for grid timings, one GPU for the neural ones.